

HYDRA Documentation

HYDRA MW4.0pe Installing AIP 8.2 on Citrix

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1 Purpose

This manual describes the necessary tasks to install the HYDRA Terminal software AIP 8.2 (Acquisition & Information Panel) on a Citrix XenApp Server.

AIP 8.2 (AIP2) is especially designed to run on industrial PCs with a touch screen monitor and without a standard keyboard and mouse connected.

Running AIP 8.2 (AIP2) on a Citrix XenApp Server instead of dedicated standard PC hardware comes along with certain restrictions, see chapter "[3 Restrictions](#)".

2 Requirements

- Citrix **XenApp 6.5**

Please note that MPDV Mikrolab GmbH does not provide any expertise regarding Citrix itself.

- Operating System: **Windows Server 2008 R2** Standard or Enterprise Edition
- It is necessary to **reboot** the Citrix (Windows) server hosting the AIP2 clients at least **every 30-40 days!** (see chapter [6.2 Automatic Restart of Terminal](#))
- One Windows user account for each running AIP2 instance.
- Unique HYDRA user ID for each running AIP2 instance (see chapter [5.2 AIP2 configuration files](#)).
The available HYDRA user ID range is **1** to **999**.
- A minimum of 500 MB capacity available on the disk drive.
- Installed Internet Explorer version **6** or newer.
- **TCP/IP** (IPv4) network configuration for the LAN/WAN network.
- Unique network ports for all installed AIP2 instances.
- The local Windows Firewall should be turned off.
- Installed **.NET Framework 3.5** (includes .NET 2.0 and 3.0).
- Installed **Visual C++ 2010 x86 Redistributable Package**.
Note: At least version 10.0.30319 of the Microsoft Visual C++ 2010 x86 Redistributable Package must be installed.
- Barcode or ID-code readers approved by MPDV Mikrolab GmbH.
To provide the full functionality of such reader systems AIP2 expects them to be connected to a local COM port.
Therefor the COM port of the Citrix client PC (e.g. Thin Client hardware) must be routed to the Citrix Server where the AIP2 software is actually running.
Those routed COM ports must be visible in the Device Manager of the Citrix server.

- Installed HYDRA 8 (MW4.0pe) server which is up and running.



This installation instruction assumes that the applications name server process

- up to and including MES Weaver 3.1: "HYDRA Nameserver"
- from MES Weaver 4.0pe onwards: "MIP Nameserver"

is listening on network port **10000**.

It is mandatory that you are checking the actual port of your HYDRA installation.

If the port of your HYDRA installation differs, you have to set a permanent environment variable in the AIP host's operating system:

```
HYPOR = ACTUAL_PORT_NUMBER
```

3 Restrictions

- Only one running AIP2 instance for each Windows user account.
Users trying to start more than one AIP2 instance inside their Citrix client session will receive an error message.
Multiple starts of AIP2 instances within the context of one user account (e.g. by parallel Citrix client and Citrix desktop sessions) must be prohibited by the Citrix administrator.
- HYDRA MDE (machine data gathering):
No automatic machine data gathering by using OPC communication, CT-UMPS, MSS, etc.
- HYDRA PDV (process data gathering):
No automatic machine data gathering by using OPC communication, CT-UMPS, MSS, etc.
- HYDRA CAQ (quality management):
No automatic measurement reading via MDI-Server (Steinwald, Ibrit, Measurement lists, etc.).
- No remote access functionality by using VNC software provided by HYDRA.

4 Pre Installation Configuration

Login as Windows user (e.g.: **Administrator**) which must be member of the local group **Administrators**.

4.1 AIP2 application directory

Decide where you want to put the AIP2 application files and create that directory.

e.g.: **C:\Program Files (x86)\MPDV\aip2**

4.2 Location for AIP2 configuration files

Each running AIP2 instance needs its own set of configuration files.

The location of those AIP2 configuration files is defined by the Windows variable **HYUSEAPPDIR**.

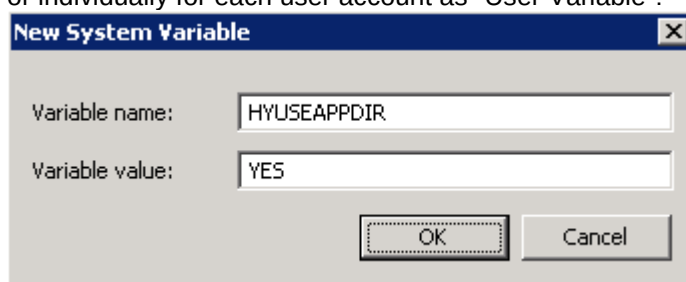
Make sure that those directories exist locally on the Citrix server.

It is mandatory to set this variable in one of the following ways:

- **HYUSEAPPDIR=YES**

either globally as "System Variable" (valid for all user accounts)

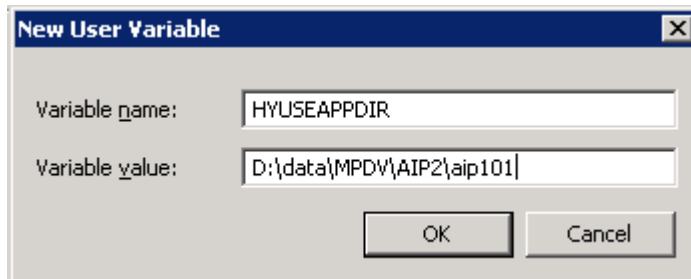
or individually for each user account as "User Variable":



AIP2 is expecting its configuration files in directory **%APPDATA%\MPDV\AIP2** of each user account (e.g.: "C:\Users\aip101\AppData\Roaming\MPDV\AIP2" for user aip101)

- **HYUSEAPPDIR=<path>**

(must be set individually for each user account as "User Variable"):



AIP2 is expecting its configuration files in **D:\data\MPDV\AIP2\aip101** if logged in with the corresponding user account.

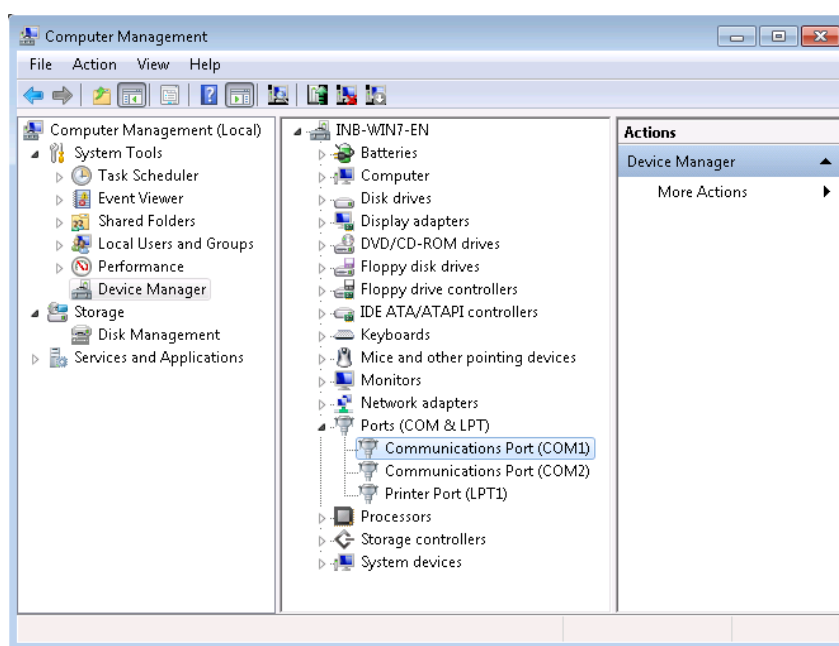
Please note that it is not recommended to use a network share path for HYUSEAPPDIR (e.g.: \\server\share) or to use a mapped network drive.

If AIP2 cannot access its configuration files permanently its proper functionality cannot be guaranteed.

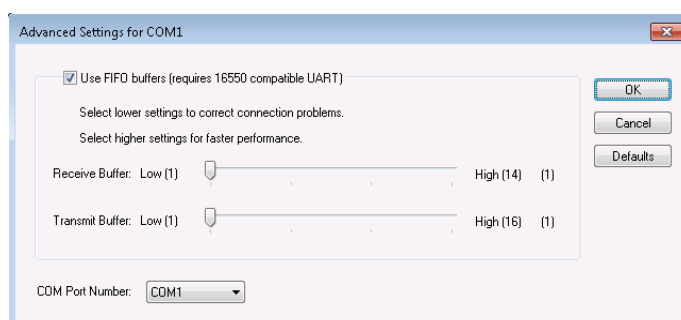
4.3 Windows Settings

- Make sure that there is one Windows user account for each running AIP2 instance available on your Citrix server.
- If you plan to use barcode or ID-code readers attached to a COM port of the Citrix client PC (e.g. Thin Client hardware) that COM port must be routed to the Citrix Server where the AIP2 software is actually running.

Those routed COM ports must be visible in the Device Manager of the Citrix server:

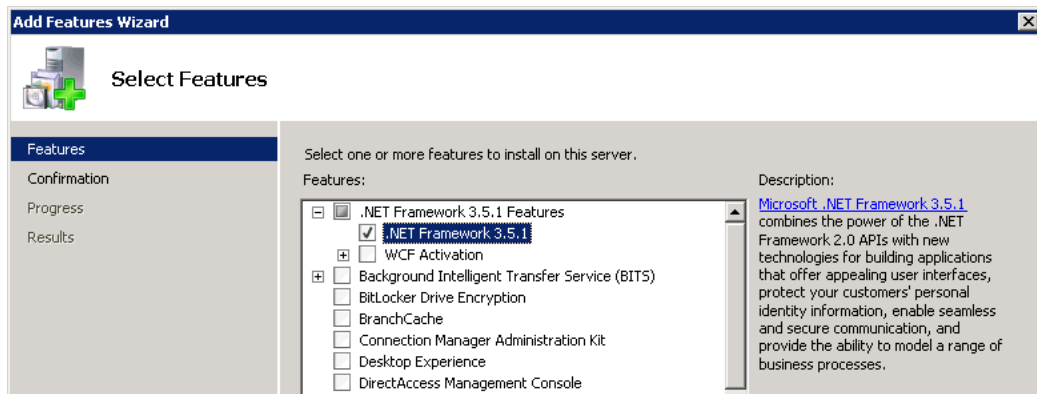


The FIFO buffer settings for all COM ports must be set to **low**:

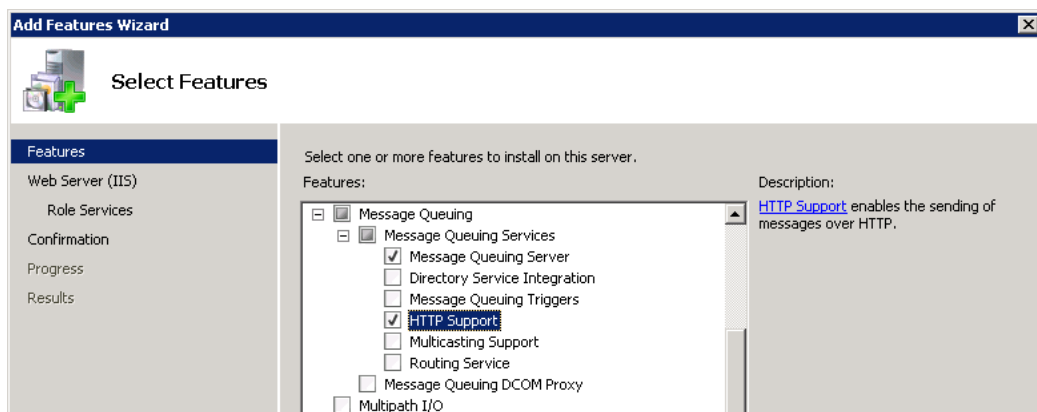


- The below-mentioned Windows configurations/components have to be installed and/or enabled to be able to use the print function for labels (List & Label):

- .Net Framework 3.5

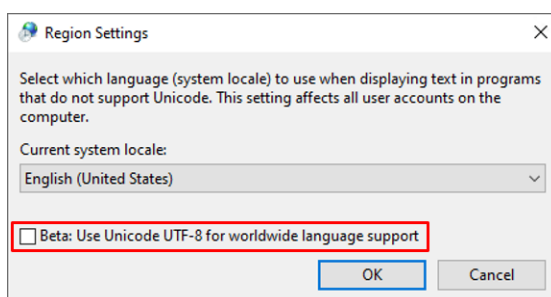


- Message-Queuing (including "MSMQ HTTP Support")



- Region Settings – Current system locale**

If there is the (Beta) checkbox "Use Unicode UTF-8 for worldwide language support" available make sure it is **not activated**:



4.4 Port Configuration

For standard AIP2 installations there will be at least the following 4 ports needed on the terminal PC:

9002 - Gateway Port for event communication (EVCOM) from the HYDRA Server to the AIP2 client. The default port 9002 can be changed in the terminal configuration set, see "Network port".

9003 - BUSPORT for communication from AIP2 to its Process Communication Controller (PCC). The default port 9003 can be changed in the AIP2 configuration file, see chapter [5.2 AIP2 configuration files](#).

9004 - BUSSERVERPORT for communication from the Process Communication Controller (PCC) to AIP2. The default port 9004 can be changed in the AIP2 configuration file, see chapter [5.2 AIP2 configuration files](#).

9005 - Gateway Port for event communication (EVCOM) from the HYDRA Server to AIP2's Process Communication Controller (PCC). The default port 9005 can be changed in the PCC configuration file, see chapter [5.3 PCC configuration files](#).

When installing multiple AIP2 terminals on a terminal server those ports need to be unique for every terminal instance.

Identify possible ports to be used for all the AIP2 instances and its Process Communication Controllers (PCC) which you want to install on your terminal server.

5 Installation

Login as Windows user (e.g.: **Administrator**) which must be member of the local group **Administrators**.

5.1 AIP2 application

- All the necessary AIP2 files are stored on the HYDRA server:

Windows: %HYDRADIR%\ctnet\win\aip2 (e.g.: d:\mip1\ctnet\win\aip2)

UNIX: \$HYDRADIR/ctnet/win/aip2 (e.g.: /u1/mip1/ctnet/win/aip2)

- Copy the following files from the above mentioned server directory to the designated AIP2 application directory on your Citrix server, e.g. to **C:\Program Files (x86)\MPDV\aip2** (see chapter [4.1 AIP2 application directory](#)):

***.exe**

***.dll**

***.mld**

***.ocx**

***.bpl**

pict.zip

- Copy the following directory (including its contents) from the above mentioned server directory to the designated AIP2 application directory on your Citrix server, e.g. to **C:\Program Files (x86)\MPDV\aip2** (see chapter [4.1 AIP2 application directory](#)):

skins (contains file **mpdv.asz**)

- If there is a customized translation file (**ctaipkd.mld**) available in the directory **%HYDRADIR%\ctnet\win\aip2\custom** on the HYDRA server it needs to be copied into the designated AIP2 application directory on your Citrix server, e.g. to **C:\Program Files (x86)\MPDV\aip2** (see chapter [4.1 AIP2 application directory](#))

- When using the print function for labels (List & Label), the ZIP file "**lprinter2.zip**" has to be extracted from the HYDRA server directory

Windows: %HYDRADIR%\ctnet\win\shared (e.g.: d:\mip1\ctnet\win\shared)

UNIX: \$HYDRADIR/ctnet/win/shared (e.g.: /u1/mip1/ctnet/win/shared)

to the AIP2 application directory.

5.2 AIP2 configuration files

- All the necessary AIP2 files are stored on the HYDRA server:
Windows: **%HYDRADIR%\ctnet\win\aip2** (e.g.: d:\mip1\ctnet\win\aip2)
UNIX: **\$HYDRADIR/ctnet/win/aip2** (e.g.: /u1/mip1/ctnet/win/aip2)
- Copy the following file from the above mentioned server directory to the designated AIP2 configuration directories on your Citrix server, e.g. to **%APPDATA%\MPDV\AIP2** (e.g.: "C:\Users\aip101\AppData\Roaming\MPDV\AIP2") (see chapter [4.2 Location for AIP2 configuration files](#)):

ctaip.bsp

- Rename file **ctaip.bsp** to **ctaip.ini**
- Open file **ctaip.ini**
- In section **[system]** enter the HYDRA user ID (**usr=**) for your designated AIP2 instance and the IP-address or the hostname of your HYDRA server (**hostname=**) and then save the file:

```
[system]
usr=101
hostname=192.168.10.101
loadfile=ctnet\win\aip2.txt
```

The available HYDRA user ID range is **1** to **999**.

If a hostname is used make sure it can be resolved to its proper IP-address (IPv4).

- In section **[DLL]** enter unique port numbers for each AIP2 instance, e.g. 8003 instead of 9003 and 8004 instead of 9004, see chapter [4.4 Port Configuration](#):

```
[Server-Communication]
...
BUSPORT=8003
...
BUSSERVERPORT=8004
```

- Make sure there is a corresponding terminal configuration set available in HYDRA for all the HYDRA user IDs you specified in the several ctaip.ini files in all the configuration directories %APPDATA%\MPDVAIP2 (see chapter [6.1 Terminal Configuration Set](#)).

5.3 PCC configuration files

- All the necessary AIP2 and PCC (Process Communication Controller) files are stored on the HYDRA server:

Windows: **%HYDRADIR%\ctnet\win\aip2** (e.g.: d:\mip1\ctnet\win\aip2)

UNIX: **\$HYDRADIR/ctnet/win/aip2** (e.g.: /u1/mip1/ctnet/win/aip2)

- Copy the following file from the above mentioned server directory to the designated AIP2 configuration directories on your Citrix server, e.g. to **%APPDATA%\MPDV\AIP2** (e.g.: "C:\Users\aip101\AppData\Roaming\MPDV\AIP2") (see chapter [4.2 Location for AIP2 configuration files](#)):

pcc.bsp

- Rename file **pcc.bsp** to **pcc.ini**
- Open file **pcc.ini**
- In section **[Server-Communication]** enter a unique port number for each AIP2's PCC instance, e.g. 8005 instead of 9005, see chapter [4.4 Port Configuration](#):

[Server-Communication]

; Communication with HYDRA server via EVCOM

Active=1

Port=8005

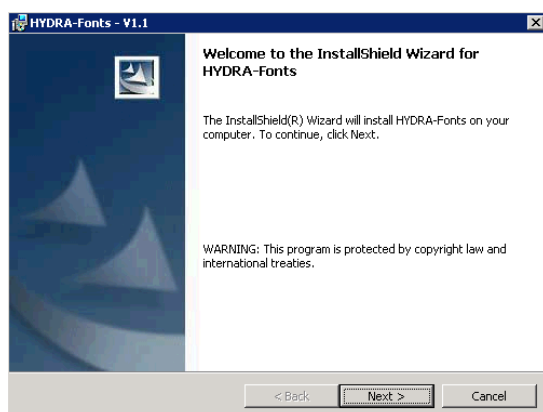
5.4 HYDRA Fonts

- Copy the installation file **hydra-fonts.msi** from the HYDRA server to a convenient directory on your Citrix server:

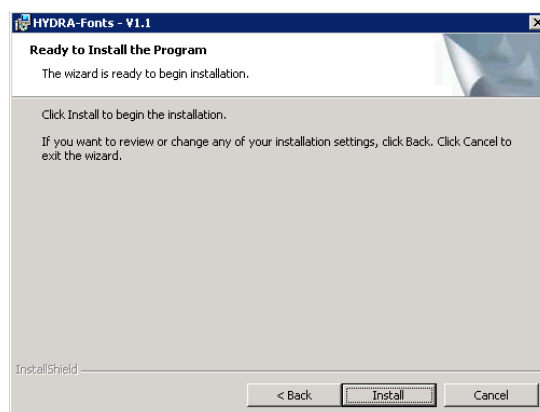
Windows: **%HYDRADIR%\ctnet\win\install\fonts\hydra-fonts.msi**

UNIX: **\$HYDRADIR/ctnet/win/install/fonts/hydra-fonts.msi**

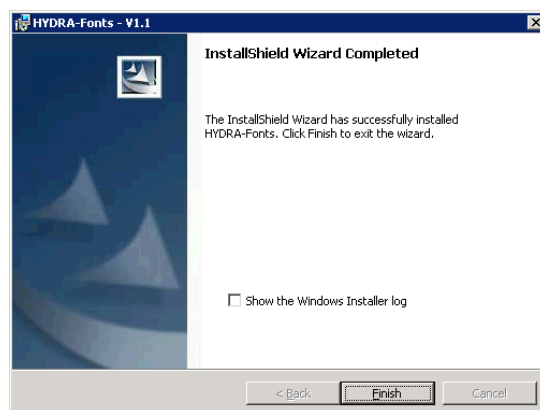
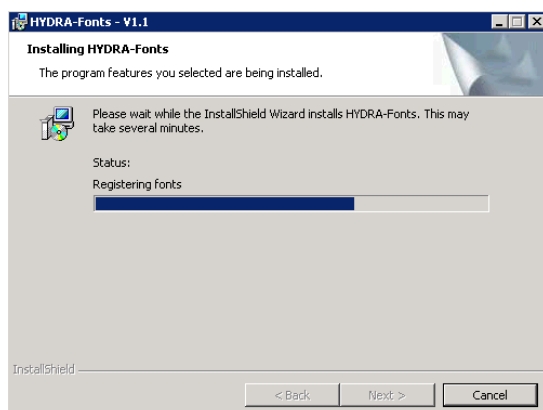
- Run the installation file **hydra-fonts.msi** on your Citrix server:



Next



Install



Finish

5.5 Publish AIP2 Software

Using the proper Citrix tools publish the AIP2 software (e.g. **C:\Program Files (x86)\MPDV\aip2\ctaip.exe**) for each specified user account so that it is available at the assigned Citrix client PCs (e.g. Thin Client hardware).

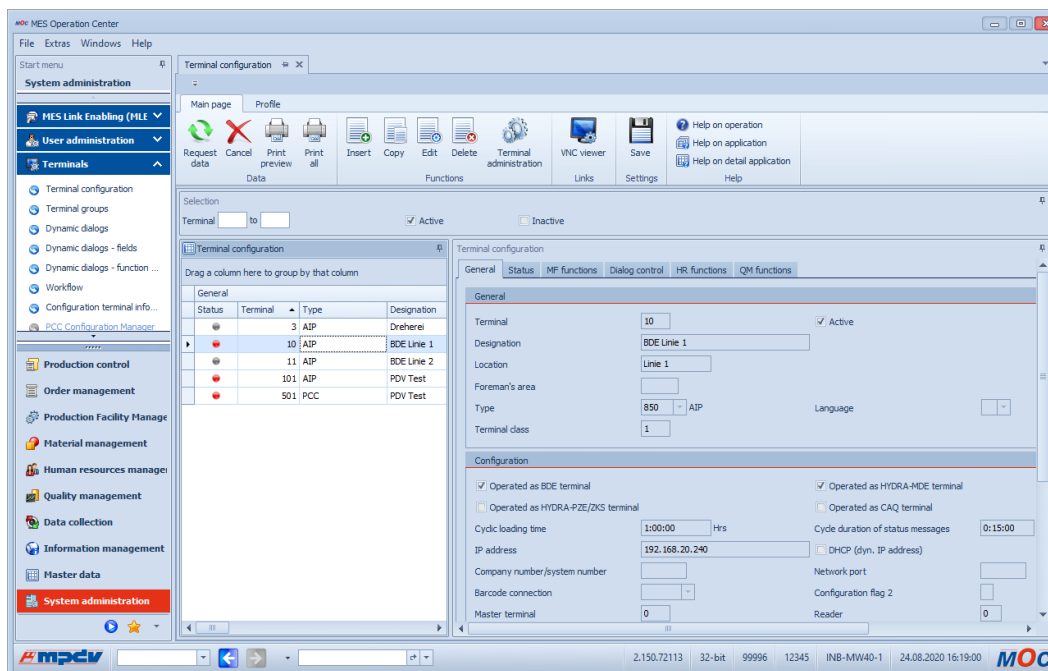
6 Post Installation Configuration

Login as Windows user (e.g.: **Administrator**) which must be member of the local group **Administrators**.

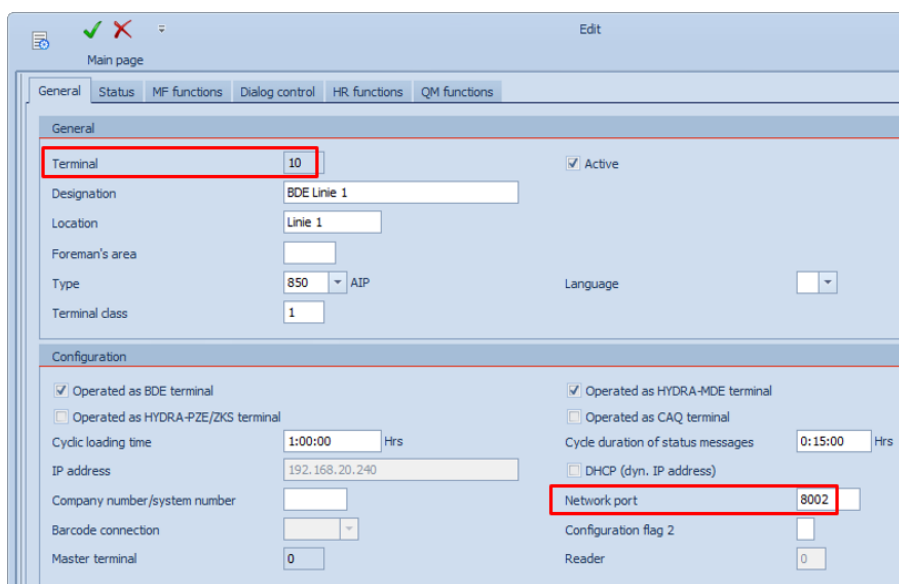
6.1 Terminal Configuration Set

Before you can use any AIP2 terminal software you have to create a terminal configuration set first by using the HYDRA client software MOC (MES Operation Center).

The terminal IDs of the configuration sets must match the HYDRA user IDs you specified in the corresponding ctaip.ini files (see chapter [5.2 AIP2 configuration files](#)).



Regarding "Network port" see chapters [4.4 Port Configuration](#) and [6.4 Gateway Communication](#).



6.2 Automatic Clock Adjustment

This configuration is mandatory for every AIP2 instance running on a Citrix XenApp server.

Deactivate the automatic clock adjustment of the HYDRA terminal software:

Add the following settings to section **[system]** in the configuration file **CTAIP.INI**:

```
[system]
NoLocalWatchUpdate=on
NoLocalWatchWarning=ON
```

By default every AIP2 client is synchronizing the local clock of the terminal PC it is installed on with the clock of the HYDRA server.

If there are additional methods enabled which are setting the clock regularly the AIP2 client will show warning messages which will prevent the use of the software as long as the message is shown.

When the automatic clock adjustment is deactivated the system administrator has to make sure that the system time of the HYDRA terminal PC (here: the Citrix server) and the HYDRA server are synchronized by other means.

6.3 Automatic Restart of Terminal

This configuration is mandatory for every AIP2 instance running on a Citrix XenApp server.

Deactivate the automatic restart for the HYDRA terminal:

Add the following settings to section **[system]** in the configuration file **CTAIP.INI**:

```
[system]
BootTermin=Off
```

There is a build-in 32bit timer in Windows which increases every one msec.

Every 49,7 days that counter will create an overflow which could cause strange behavior of the AIP2 client software like hang-ups.

Therefore the HYDRA terminal software will reboot the terminal PC automatically on the first coming Sunday at 1:30am after a 30 day period.

ATTENTION:

The Citrix administrator has to make sure to reboot the Citrix (Windows) server hosting AIP2 clients before that 49,7 days period elapses.

It is recommended to reboot the Citrix (Windows) server at least every 30–40 days!

6.4 Gateway Communication AIP2

This configuration is mandatory for every AIP2 instance running on a Citrix XenApp server.

AIP2's gateway communication (EVCOM) will be used by PDV functionality which is not supported on terminal server installations.

Therefore you might want to deactivate the gateway communication option for all AIP2 instances.

Add the following settings to section **[GateWay-Communication]** in the configuration file **CTAIP.INI** (create that section if it does not exist):

```
[GateWay-Communication]
Active=false
```

By default the gateway communication option for the HYDRA terminal is active and will use port 9002 as default.

On a terminal server environment running more than one AIP2 client it is only possible for one of those clients to acquire that port.

All others will show an error message during their start up when using the default configuration.

If the gateway communication (EVCOM) should be used for any reasons then it is mandatory to assign a **unique port** to every AIP2 instance. Add the following settings to section **[GateWay-Communication]** in the configuration file **CTAIP.INI**:

```
[GateWay-Communication]
Active=true
```

Make sure that a **unique port** is defined in the corresponding terminal configuration set of every AIP2 instance, e.g. 8002 instead of 9002 (see chapters [4.4 Port Configuration](#) and [6.1 Terminal Configuration Set](#)):

Network port	8002
--------------	------

6.5 Gateway Communication PCC

This configuration is mandatory for every AIP2 instance running on a Citrix XenApp server.

The gateway communication (EVCOM) of AIP2's Process Communication Controller (PCC) will be used by PDV functionality which is not supported on terminal server installations.

Therefore you might want to deactivate the gateway communication option for all PCC instances.

Change the following settings in section **[Server-Communication]** in the PCC configuration file **PCC.INI**:

```
[Server-Communication]
; Communication with HYDRA server via EVCOM
Active=0
Port=9005
```

By default the gateway communication option for the PCC is active and will use port 9005 as default.

On a terminal server environment running more than one AIP2 client it is only possible for one of their PCCs to acquire that port.

All others will show an error messages in their log files when using the default configuration.

If the gateway communication (EVCOM) should be used for any reasons then it is mandatory to assign a **unique port** to every AIP2's PCC instance. Change the following settings in section **[Server-Communication]** in the PCC configuration file **PCC.INI**:

```
[Server-Communication]
; Communication with HYDRA server via EVCOM
Active=1
Port=8005
```

See chapters [4.4 Port Configuration](#) and [5.3 PCC configuration files](#).

6.6 Barcode or ID-Code Readers

This configuration is mandatory for every AIP2 instance which is using barcode or ID-code readers connected to a COM port.

Depending on which COM port the assigned reader hardware is connected add the appropriate settings to section **[comports]** in the configuration file **CTAIP.INI**.

e.g.:

```
[comports]
com5=BAR
com6=LEGIC
```

By default AIP2 is scanning for up to 30 COM ports.

If there are more than 30 COM ports available at the Citrix server you have to add the COMCOUNT setting to section **[comports]** in the configuration file **CTAIP.INI**.

The value for COMCOUNT depends on the available amount of COM ports at the server.

e.g.:

```
[comports]
COMCOUNT=60
com45=BAR
com46=LEGIC
```

6.7 Operating without Touch Screen Monitor

AIP2 is especially designed to run on industrial PCs with a touch screen monitor and without a standard keyboard and mouse connected.

Therefore AIP2 is hiding the Windows mouse cursor by default.

If the Citrix client PC (e.g. Thin Client hardware) does not provide a touch screen monitor you need to tell AIP2 not to hide the Windows mouse cursor.

That will improve the usability on such an environment.

Add the following settings to section **[system]** in the configuration file **CTAIP.INI**:

```
[system]
showcursor=ON
```

6.8 Restricting AIP2 Window Size

By default AIP2 is using the available screen resolution provided by the Windows operating system so its window size will use up the whole screen.

If it should be necessary to restrict the AIP2 window size add the following settings to section **[system]** in the configuration file **CTAIP.INI**:

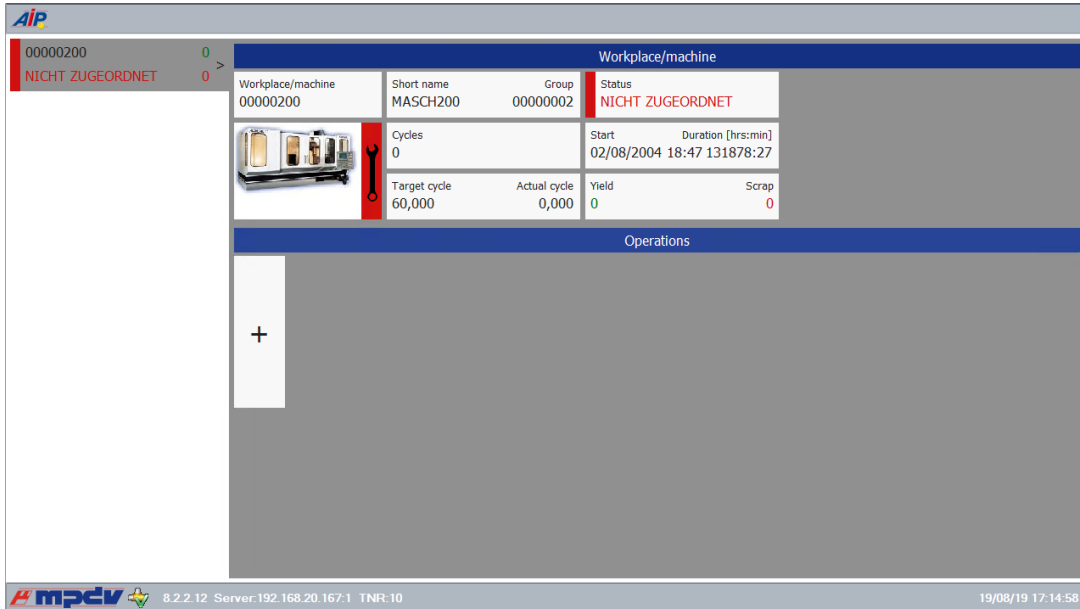
e.g.:

```
[system]
VirtScreenSize=1280
```

7 Ready to Go

The basic installation of the terminal software is finished now.

The terminal(s) can be started now.



The screenshot displays the AIP 8.2 software interface. On the left is a sidebar with a tree view containing a red bar with the text 'NICHT ZUGEORDNET' and a plus sign. The main area shows a table with the following data:

Workplace/machine			
Workplace/machine 00000200	Short name MASCH200	Group 00000002	Status NICHT ZUGEORDNET
	Cycles 0	Start 02/08/2004 18:47	Duration [hrs:min] 131878:27
	Target cycle 60,000	Actual cycle 0,000	Yield 0
Scrap 0			

Below the table is a section titled 'Operations' with a large grey area and a plus sign on the left. The bottom status bar shows the mpcv logo, version 8.2.2.12, server information (192.168.20.167:1 TNR:10), and the date/time 19/08/19 17:14:58.

8 Updating AIP2 Software

Login as Windows user (e.g.: **Administrator**) which must be member of the local group **Administrators**.

Shutdown all running AIP2 instances!

All the necessary AIP2 files are stored on the HYDRA server:

Windows: **%HYDRADIR%\ctnet\win\aip2** (e.g.: d:\mip1\ctnet\win\aip2)

UNIX: **\$HYDRADIR/ctnet/win/aip2** (e.g.: /u1/mip1/ctnet/win/aip2)

Copy the following files from the above mentioned server directory to the designated AIP2 application directory on your Citrix server, e.g. to **C:\Program Files (x86)\MPDV\aip2** (see chapter [4.1 AIP2 application directory](#)):

***.exe**
***.dll**
***.mld**
***.ocx**
***.bpl**
pict.zip

Replace all existing files.

Copy the following directory (including contents) from the above mentioned server directory to the designated AIP2 application directory on your Citrix server, e.g. to **C:\Program Files (x86)\MPDV\aip2** (see chapter [4.1 AIP2 application directory](#)):

skins (contains file **mpdv.asz**)

Replace all existing files.

When using the print function for labels (List & Label), the ZIP file "**lprinter2.zip**" has to be extracted from the HYDRA server directory

Windows: **%HYDRADIR%\ctnet\win\shared** (e.g.: d:\mip1\ctnet\win\shared)

UNIX: **\$HYDRADIR/ctnet/win/shared** (e.g.: /u1/mip1/ctnet/win/shared)

to the AIP2 application directory where all existing files are replaced.

Start your AIP2 instances.